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ANRF AISEHack 2026

Finale presentation | Theme : Flood Detection

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Salient Contribution:
- **Best Score: Flood IoU = 0.1981**
- **ResNet-101 + UNet++ with learnable CBAM Channel Adapter**

Our salient contributions

f1

Modelling strategy

ResNet-101 encoder + UNet++ decoder with custom CBAM Channel Adapter
- 5-band SAR+Optical input (HH, HV, Green, Red, NIR)
- CBAM adapter projects 5ch → 3ch preserving ImageNet pretrained weights
- Focal Loss + Dice Loss with sweep-optimized class weights [0.15, 0.60, 0.25]

f2

Training

- Freeze backbone 10 epochs → Unfreeze → CosineAnnealing LR (1e-4 → 1e-6)
- Gradient accumulation: batch 4 × accum 4 = effective batch 16
- Augmentations: HFlip, VFlip, Rotate90, GaussNoise
- Early stopping on Flood IoU (patience=15)
- Systematic alpha sweep + band ablation to find optimal config

f3

Results + Actions

- Best Leaderboard Score: Flood IoU = 0.1981
- Key insight: Dropping SWIR band + tuned focal weights gave biggest gains
- Flood-Priority Override at inference: force flood where P(flood) > threshold
- 9 alternative strategies tested and documented (negative results)

Strategy	Description	Result	Why It Failed
HRNet-W48	High-Resolution Network encoder to capture fine-grained spatial features	Flood IoU < baseline	The small dataset (59 images) could not support the model's larger capacity; severe overfitting
EfficientNet-B7 + UNet++	Heavier encoder with compound scaling	Matched baseline, did not exceed	Marginal gains consumed by overfitting on the small dataset
Test-Time Augmentation (TTA)	8-fold geometric augmentation at inference	Negligible improvement (~0.002)	Water body boundaries are rotation-invariant; TTA added noise rather than signal
Early Fusion of Auxiliary Data	Stacking JRC + CartoDEM + Slope as 3 extra input channels (8ch total)	Flood IoU dropped to 0.1321	The channel adapter could not learn to jointly process spectral reflectance, elevation (meters), and water probability (%) in a single convolutional pass
BiRefNet (Unfrozen)	Fully unfreezing the BiRefNet backbone for fine-tuning	Catastrophic performance degradation	Pretrained weights were destroyed on the tiny dataset
CutMix Augmentation	Cut-and-paste augmentation between training images	Flood IoU = 0.1493	Introduced artificial flood/water boundaries that confused the decoder
Lovasz-Softmax Loss	Direct IoU optimization as a loss function	No improvement over FocalDice	Unstable gradients with the small batch sizes we used
Model Soups	Averaging weights of models trained with different seeds	Did not complete in time	GPU bottleneck on DataLoader I/O; each ResNet-101 training run took ~5 hours
6-Band Input (with SWIR)	Using all 6 satellite bands including SWIR	Worse than 5-band	SWIR introduced noise in flood boundary regions

Notebook editor cells

: Input Band Selection

Rank	experiment	in_channels	best_epoch	val_mean_iou	val_iou_flood	val_iou_water	val_loss
1	all_minus_swir	5	21	0.445698	0.175483	0.416142	0.071703
2	sar_nir_swir	4	24	0.441812	0.164803	0.394575	0.080441
3	sar_only	2	30	0.425395	0.173226	0.356001	0.086703
4	all_6bands	6	20	0.421659	0.141673	0.365424	0.076950
5	optical_only	4	27	0.320704	0.038649	0.293751	0.103652

🏆 WINNER: all_minus_swir
mIoU=0.4457 | Flood IoU=0.1755

MATCHMAKING

Modelling - *Strategies*

01

Custom learnable adapter for multi-spectral input

- Satellite data has 5 bands (HH, HV, Green, Red, NIR)
- ImageNet encoders expect 3 channels (RGB)
- CBAM adapter: 5ch → Conv(32ch) → Channel Attention → Conv(3ch)
- Learns which bands matter most (SAR vs Optical)

01

02

ResNet-101 (ImageNet pretrained) + UNet++ Decoder

- Encoder frozen for 10 epochs, then unfrozen
- Decoder channels: (256, 128, 64, 32, 16)
- Output: 3-class segmentation (Land, Flood, Permanent Water)

02

03

Focal Loss + Dice Loss with systematically tuned weights

- 7 focal alpha configs swept → best: [0.15, 0.60, 0.25]
- Flood class weighted 4x higher than Land
- Band ablation: tested all 6-band combos → 5 bands (drop SWIR) wins

03

Architecture diagram

Input (5 bands: HH, HV, G, R, NIR)

↓

CBAM Adapter (5ch → 3ch)

↓

ResNet-101 Encoder (frozen → unfrozen)

↓

UNet++ Decoder

↓

3-Class Output (Land | Flood | Water)

↓

Flood-Priority Override (post-processing)

Strengths:
Preserves ImageNet features; systematic sweeps found optimal config on tiny dataset

Weaknesses:
59 training images → overfitting; cannot spectrally distinguish flood from permanent water

Key Results – *Why , what and how of the workings?*

S1

What is working?

Architecture: ResNet-101 + UNet++ + CBAM Adapter (5ch input)
Training: Freeze 10 epochs → Unfreeze, CosineAnnealing LR, grad accum (eff. batch=16)
Data: 5 bands (HH, HV, G, R, NIR), Z-score normalization, HFlip/VFlip/Rotate90/GaussNoise
Key hyperparams: focal_alpha=[0.15, 0.60, 0.25], LR=1e-4, patience=15

● Implemented

All core setup criteria done

S2

How is it working? (Quantitative evals)

Test mIoU: 0.4926 | Flood IoU: 0.1779 | Water IoU: 0.5212 | Accuracy: 74.55%
Leaderboard (blind 19 images): Flood IoU = 0.1981
Alpha sweep [0.15,0.60,0.25] + dropping SWIR band → +0.02 gain over baseline (0.1773)

S3

Why is it working? (Reasoning and analysis)

1. CBAM adapter learns SAR bands are most important for water detection
2. Dropping SWIR removes noisy flood boundary pixels
3. Focal alpha [0.15, 0.60, 0.25] weights flood class 4x over land
4. Flood-Priority Override catches borderline flood pixels at inference
5. Simple well-tuned model beats complex architectures on 59 images

S4

Params: ~60M | Accelerator: Tesla T4 (15GB) | Training: ~15 min | Inference: ~0.3s/image

Phase 1 → Phase 2:
- Phase 1: 5-class, basic UNet, no adapter, no sweeps
- Phase 2: 3-class, CBAM adapter, alpha sweep, band ablation, strict seeding, Flood-Priority Override → 0.1981

MATCHMAKING

Key Results - *Why , what and how of the workings?*

01

Alpha Sweep Results:
alpha=[0.15, 0.60, 0.25] → 0.1981 (BEST)
alpha=[0.20, 0.50, 0.30] → 0.1773
alpha=[0.10, 0.70, 0.20] → lower
Finding: Flood weight of 0.60 is the sweet spot

01

02

Band Ablation Results:

5 bands (HH,HV,G,R,NIR) → 0.1981 (BEST)
6 bands (+ SWIR) → lower

Finding: SWIR adds noise at flood boundaries

02

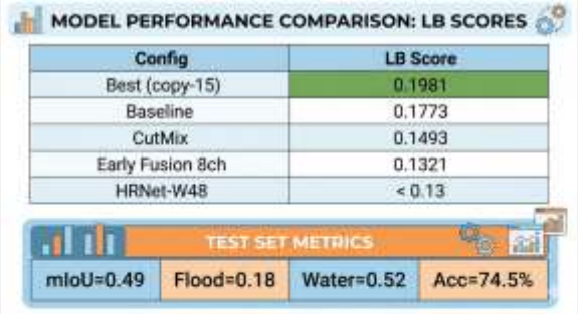
03

Reproducibility & Deterministic Training:
- Strict seeding: CUBLAS_WORKSPACE_CONFIG, per-worker DataLoader seeds
- CosineAnnealing LR (1e-4 → 1e-6) + Early Stopping (patience=15)
- Freeze backbone 10 epochs → Unfreeze → fine-tune full model

Finding: Deterministic setup enabled reliable A/B comparison across experiments

03

Plots/Tables


MODEL PERFORMANCE COMPARISON: LB SCORES

Config	LB Score
Best (copy-15)	0.1981
Baseline	0.1773
CutMix	0.1493
Early Fusion 8ch	0.1321
HRNet-W48	< 0.13

TEST SET METRICS

mIoU=0.49	Flood=0.18	Water=0.52	Acc=74.5%

Strengths: Systematic sweeps found optimal config; reproducible; only 15 min training
Weaknesses: 59 images → overfitting; flood and water spectrally identical

MATCHMAKING

Details of Relevant Experimentation and Methodology

wX4f17TdY0hWsrjxNpHh6OwjtNK1nUdb03x
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01

- Band ablation: 5 bands (drop SWIR) > 6 bands
- Alpha sweep: 7 configs tested → [0.15, 0.60, 0.25] best
- Failed: JRC post-processing, Early Fusion (8ch), BiRefNet, HRNet-W48, CutMix, Model Soups, TTA — all reduced score

01

02

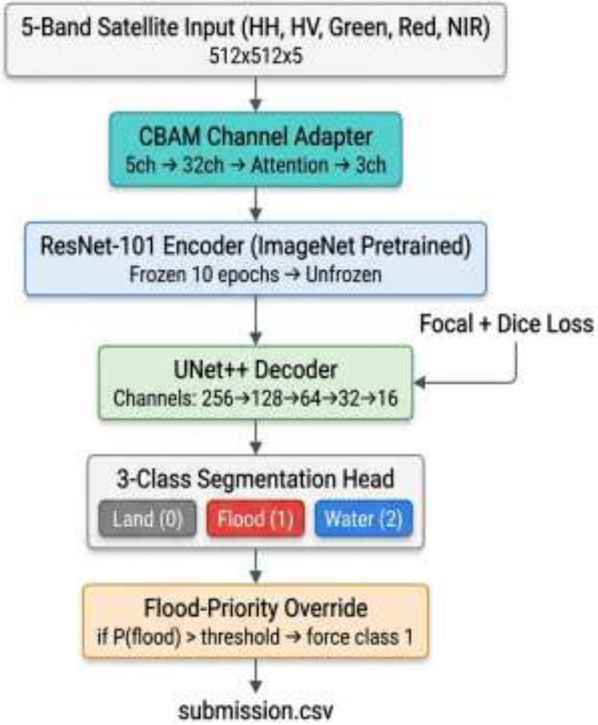
- Segmentation Models PyTorch (smp) — Yakubovskiy, 2019
- CBAM: Convolutional Block Attention Module — Woo et al., ECCV 2018
- Model Soups — Wortsman et al., ICML 2022 (attempted, did not finish)
- JRC Global Surface Water — Pekel et al., Nature 2016 (explored, did not improve)

02

03

- Used AI coding assistant for architecture design, debugging, sweep automation
- Key AI contributions: CBAM adapter design, alpha sweep notebook, reproducibility fixes (seeding), band ablation framework
- All strategy decisions (model choice, final config) made by team
- Full transcript: genai_transcript.md in GitHub repo

03



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